



Question Set

Question 1

Consider an implied volatility smile and surface used to price exotic options. Define the state variables (log-moneyness, total variance, expiry, local volatility, and smile parameters) and derive the SVI total variance and Dupire local-volatility relation. State the measure, filtration, and tradability assumptions.

$$w(k) = a + b\{\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2}\}$$

Question 2

For an implied volatility smile and surface used to price exotic options, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in an implied volatility smile and surface used to price exotic options. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using five strikes across log-moneyness from -0.35 to 0.35 and expiries from 3M to 3Y, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

$$w(k) = a + b\{\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2}\}$$

Question 5

For an implied volatility smile and surface used to price exotic options, compute and interpret smile vega, skew risk, and volga. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes SVI or SABR parameters fit to implied vol quotes. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for an implied volatility smile and surface used to price exotic options. Explain how no-arbitrage positivity and wing behavior prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for an implied volatility smile and surface used to price exotic options. Specify surface-implied scenario shocks to volatility, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements an implied volatility smile and surface used to price exotic options and gets a stable but wrong answer. Explain how fitting total variance at one expiry but comparing raw vol across expiries creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for an implied volatility smile and surface used to price exotic options. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of an implied volatility smile and surface used to price exotic options, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of five strikes across log-moneyness from -0.35 to 0.35 and expiries from 3M to 3Y. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for an implied volatility smile and surface used to price exotic options. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in an implied volatility smile and surface used to price exotic options. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for an implied volatility smile and surface used to price exotic options is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for SVI or SABR parameters fit to implied vol quotes are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For an implied volatility smile and surface used to price exotic options, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for surface-implied scenario shocks to volatility. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that fitting total variance at one expiry but comparing raw vol across expiries could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain an implied volatility smile and surface used to price exotic options and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.